Loss of rapid velocity decay in a broad smooth class for the cutoff Boltzmann equation

Kaiwen Jin
yui.ui.siki@gmail.com

15 October 2026

Abstract

We construct a spatially homogeneous, radial, nonnegative smooth solution of the full-sphere hard-sphere Boltzmann equation whose initial value is a Gaussian but whose velocity tail is a positive multiple of r^{-6} at every positive time. Mass is one, and energy and entropy have uniform finite upper bounds on a short time interval. This gives a negative answer to a literal rapid-decay continuation statement when smoothness means joint pointwise smoothness and no energy admissibility or positive-time Schwartz condition is part of the solution class. The solution itself continues smoothly in the critical weighted class: the example concerns loss of rapid velocity decay and does not establish local derivative blowup or resolve conservative continuation. The proof separates an explicit radial linear part from a nonlinear remainder with one extra power of decay, and proves nonnegativity by a compact-maximum argument. No novelty claim is made.

1 Statement and interpretation

Consider

$$\partial_t f + v \cdot \nabla_x f = Q(f,f), \qquad Q(F,F)(v) = \int_{\mathbb{R}^3}\int_{\mathbb{S}^2} |v - v_*|[F(v')F(v'_*) - F(v)F(v_*)]\, d\sigma\, dv_*, \tag{1.1}$$

where

$$v' = \frac{v+v_*}{2} + \frac{|v-v_*|}{2}\sigma, \qquad v'_* = \frac{v+v_*}{2} - \frac{|v-v_*|}{2}\sigma.$$

The sphere has its usual surface measure of total area 4π; $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$. All collision factors in (1.1) have the same t, x.

The formulation of AIM problem 366 asks whether a nonnegative smooth solution on $[0,T)\times\mathbb{T}^3\times\mathbb{R}^3$, initially rapidly decreasing together with all derivatives, extends beyond T with the same rapid velocity decay if, uniformly in t, x,

$$m_0 \le \int_{\mathbb{R}^3} f\, dv \le M_0, \qquad \int_{\mathbb{R}^3} |v|^2 f\, dv \le E_0, \qquad \int_{\mathbb{R}^3} f\log f\, dv \le H_0. \tag{1.2}$$

Here $m_0, M_0, E_0 > 0$, $H_0 \in \mathbb{R}$, and $0\log 0 = 0$. We address the literal reading in which smoothness is joint pointwise C^∞ regularity. In particular, no energy inequality or locally uniform positive-time Schwartz membership is assumed. The distinction between these solution classes is essential to the conclusion.

For the time regularity assertion, let X_6 be the space of continuous radial profiles $F : [0,\infty) \to \mathbb{R}$ with

$$\|F\|_6 = \sup_{r\ge 0}(1+r^2)^3|F(r)| < \infty.$$

A homogeneous radial function is identified with its profile in this space.

Theorem 1.1. *There exist $\varepsilon > 0$, $t_* > 0$, and a nonnegative spatially homogeneous radial solution f of (1.1), jointly C^∞ on $[0, t_*] \times \mathbb{T}^3 \times \mathbb{R}^3$, such that*

$$f(0, x, v) = \pi^{-3/2} e^{-|v|^2}, \qquad \int_{\mathbb{R}^3} f(t, x, v)\, dv = 1, \tag{1.3}$$

$$\sup_{t,x} \int_{\mathbb{R}^3} |v|^2 f(t, x, v)\, dv < \infty, \qquad \sup_{t,x} \int_{\mathbb{R}^3} f(t, x, v) \log f(t, x, v)\, dv < \infty, \tag{1.4}$$

$$\lim_{r\to\infty} r^6 f(t, x, r\omega) = 4\varepsilon e^{-1/t^2} > 0 \quad (0 < t \le t_*,\ x \in \mathbb{T}^3,\ \omega \in \mathbb{S}^2). \tag{1.5}$$

All spatial and velocity derivatives of the initial datum are Schwartz. The solution is C^∞ in time with values in X_6.

Smoothness at a closed time endpoint means smooth one-sided traces. The final t_* will lie strictly inside the interval on which the solution is constructed. Thus the example already has a smooth continuation in X_6. For every $0 < T < t_*$, its restriction to $[0, T)$ satisfies (1.2), but its endpoint at T cannot be rapidly decreasing. No conclusion about blowup of local derivatives, or about the conservative or energy-admissible continuation problem, follows from this example.

The cutoff regularity question is discussed in Silvestre's survey [1, Section 8.5]. The formulation used here is the explicit hard-sphere statement of AIM problem 366.[1] We use the account in Basile et al. [3, Sections 1 and 7] for the energy-increasing weak-solution precedent and its attribution to Lu and Wennberg [2]. Basile et al. also give a weak construction. Our reference to Lu–Wennberg follows that account; their full theorem statements remain unverified here. The account provides context for contributions from arbitrarily large velocities. We do not use a kinetic existence theorem from these works, and make no novelty or priority claim.

The proof uses the leading radial operator

$$C_0 F(r) = \frac{4}{r} \int_r^\infty sF(s)\, ds - rF(r).$$

Its r^{-6} tail is critical: the two displayed terms cancel on that power. After a cutoff and a mass correction, the difference between the nonlinear collision operator and this leading part decays one power faster. We prescribe a flat positive tail, solve for a faster-decaying correction, and finally prove nonnegativity. Sections 2–5 give these steps.

2 Radial identities and leading terms

Write $r = |v|$, $e = r^2$, $F(r) = F(v)$, and $G(e) = F(\sqrt{e})$. We allow signed radial profiles until positivity has been proved. Define

$$m(F) = 4\pi \int_0^\infty r^2 F(r)\, dr, \qquad \nu_F(v) = 4\pi \int_{\mathbb{R}^3} |v - w| F(w)\, dw,$$

and let $Q^+(F_1, F_2)$ denote the gain integral with post-collision factors $F_1(v')F_2(v'_*)$. It is a symmetric bilinear map whenever the integral is absolute; symmetry follows by replacing σ with $-\sigma$. Then $Q(F, F) = Q^+(F, F) - \nu_F F$.

For $s > 0$, put $g_s(r) = \langle r\rangle^{-s}$, where $\langle r\rangle = (1 + r^2)^{1/2}$, and set

$$X_s = \{F \in C([0, \infty)) : \|F\|_s := \sup_{r\ge 0} \langle r\rangle^s |F(r)| < \infty\}.$$

[1] We use this archived version of the problem statement. The accompanying target comparison records its correspondence with the note. Its interpretation, rather than a claim to resolve every conventional formulation of conditional regularity, is the point of the example.

Let X_s^0 be the closed subspace on which $\langle r\rangle^s F(r) \to 0$. Multiplication by $\langle r\rangle^s$ identifies X_s isometrically with the bounded continuous functions, so these spaces are Banach spaces. Compactly supported continuous profiles are dense in X_s^0. Fix throughout

$$p = \frac{13}{2}, \qquad 6 < p < 7, \qquad M(r) = \pi^{-3/2} e^{-r^2}. \tag{2.1}$$

The Gaussian has mass one and satisfies $Q(M, M) = 0$ by conservation of the sum of the two squared speeds in a single collision. This identity does not assert an energy balance for a time-dependent solution. For $j = 0, 1, 2$,

$$\int_{\mathbb{R}^3} |v|^j |F(v)|\, dv \le 4\pi \|F\|_6 \int_0^\infty r^{j+2}(1+r^2)^{-3}\, dr < \infty.$$

Thus mass and the first two absolute moments are bounded on X_6. So are $F \mapsto \int G(e)\, de$ and $F \mapsto \int eG(e)\, de$.

The exact collision formulas

For $e > 0$, define the tail and local integrals

$$A_e = \int_e^\infty G(s)\, ds, \qquad B_e = \int_0^e \sqrt{s}\, G(s)\, ds,$$
$$D_e = \int_e^\infty sG(s)\, ds, \qquad E_e = \int_0^e s^{3/2} G(s)\, ds,$$
$$C_e = \int_0^e ds \int_{e-s}^e dt\, \sqrt{s+t-e}\, G(s)G(t).$$

For every radial $F \in X_6$,

$$\frac{Q^+(F,F)(r)}{8\pi^2} = A_e^2 + \frac{2B_eA_e}{r} + \frac{C_e}{r}, \tag{2.2}$$
$$\frac{\nu_F(r)}{8\pi^2} = rB_e + \frac{E_e}{3r} + D_e + \frac{e}{3}A_e. \tag{2.3}$$

Here and below the formulas at $r = 0$ mean their continuous limits.

To check the normalization, the unsplit radial formula is

$$\frac{Q(F,F)(r)}{8\pi^2} = \frac{1}{r}\int_0^\infty de_* \int_0^{e+e_*} de_1\, \min(\sqrt{e}, \sqrt{e_*}, \sqrt{e_1}, \sqrt{e_2})[G(e_1)G(e_2) - G(e)G(e_*)], \tag{2.4}$$

where $e_2 = e + e_* - e_1$. Indeed, for $s = |v_*|$, let z be the cosine between v and v_*, $U = |v + v_*|$, and $d = |v - v_*|$. If θ is the angle between $v + v_*$ and σ, then

$$e_1 = \frac{r^2 + s^2 + Ud\cos\theta}{2}.$$

Thus e_1 is uniform on an interval of length Ud. The two azimuthal integrations give $8\pi^2\ dz\ de_1/(Ud)$. Multiplication by the hard-sphere factor d and use of $dz = U\ dU/(rs)$ leave $8\pi^2\ dU\ de_1/(rs)$. For fixed output speeds the allowed U-interval is

$$[|r-s|, r+s] \cap [|\sqrt{e_1} - \sqrt{e_2}|, \sqrt{e_1} + \sqrt{e_2}],$$

of length $2\min(r, s, \sqrt{e_1}, \sqrt{e_2})$. Finally $s\ ds = de_*/2$ gives (2.4).

For the gain, change variables from (e_*, e_1) to $(\xi, \zeta) = (e_1, e_2)$. The Jacobian is one, $e_* = \xi + \zeta - e$, and the domain is $\xi, \zeta \ge 0$, $\xi + \zeta \ge e$. When both output energies exceed e,

the minimum kernel is $\sqrt{e} = r$; after division by r this region contributes A_e^2. If $\xi \le e \le \zeta$, the minimum is $\sqrt{\xi}$, and the contribution is $B_e A_e / r$. The exchanged region contributes the same amount. In the remaining triangle $\xi, \zeta \le e$, $\xi + \zeta \ge e$, the minimum is $\sqrt{\xi + \zeta - e}$, giving C_e / r. These three contributions prove (2.2).

For the loss, direct integration of the same kernel gives

$$\int_0^{e+e_*} \min(\sqrt{e}, \sqrt{e_*}, \sqrt{y}, \sqrt{e + e_* - y})\, dy = \sqrt{\min(e, e_*)} \left[\max(e, e_*) + \frac{\min(e, e_*)}{3} \right].$$

Indeed, split y at $\min(e, e_*)$ and $\max(e, e_*)$. The two end integrals give $4 \min(e, e_*)^{3/2}/3$; the middle one is $\sqrt{\min(e, e_*)}\, |e - e_*|$. After division by r, the regions $e_* \le e$ and $e_* \ge e$ give respectively $r B_e + E_e/(3r)$ and $D_e + e A_e / 3$. This proves (2.3). All coordinate changes can first be made for nonnegative profiles. The estimates below give finite majorants for the g_6 envelope, and hence justify the signed formulas by absolute integration.

There is no singularity at zero. Directly in the original operator,

$$Q^+(F, F)(0) = 8\pi^2 \left(\int_0^\infty G(s)\, ds \right)^2, \qquad \nu_F(0) = 8\pi^2 \int_0^\infty s G(s)\, ds.$$

For later smoothness arguments, the local terms have the useful forms

$$\begin{aligned} \frac{B_e}{r} &= e \int_0^1 \sqrt{b}\, G(eb)\, db, & r B_e &= e^2 \int_0^1 \sqrt{b}\, G(eb)\, db, \\ \frac{E_e}{3r} &= \frac{e^2}{3} \int_0^1 b^{3/2} G(eb)\, db, & \frac{C_e}{r} &= e^2 \int_0^1 db \int_{1-b}^1 dd\, \sqrt{b + d - 1}\, G(eb) G(ed). \end{aligned} \tag{2.5}$$

They vanish to the stated orders at zero.

The mass correction and the faster remainder

Choose a fixed smooth $0 \le \chi \le 1$, zero on $[0, 4]$ and one on $[8, \infty)$. For example, with $\eta(u) = e^{-1/u}$ for $u > 0$ and zero otherwise, take $\chi(r) = \eta(r - 4)/[\eta(r - 4) + \eta(8 - r)]$. Recall that M is the unit-mass Gaussian in (2.1). For $r > 0$, the leading expression is

$$C_0 F(r) = \frac{4}{r} \int_r^\infty s F(s)\, ds - r F(r).$$

Define

$$C_{\mathrm{raw}} F = \chi C_0 F, \qquad \ell(F) = m(C_{\mathrm{raw}} F), \qquad CF = C_{\mathrm{raw}} F - \ell(F) M, \qquad \mathcal{L} F = 4\pi C F. \tag{2.6}$$

The raw expression is zero near zero. For $s > 4$, ℓ is bounded on X_s: the loss mass is controlled by $\int_4^\infty r^3 |F(r)|\, dr$, and Tonelli gives the same bound for the gain mass. In particular $m(CF) = 0$. The expression $\mathcal{L} F$ is defined pointwise on X_s; it need not belong to X_s.

Lemma 2.1. *The map*

$$R(F) = Q(F, F) - 4\pi m(F) C F$$

is a continuous quadratic map $X_6 \to X_7$, with

$$\begin{aligned} \|R(F)\|_7 &\le C \|F\|_6^2, \\ \|R(F) - R(\widetilde{F})\|_7 &\le C(\|F\|_6 + \|\widetilde{F}\|_6) \|F - \widetilde{F}\|_6. \end{aligned} \tag{2.7}$$

It has zero mass. Since $p < 7$, its values belong to X_p^0. On the affine space $m(F) = 1$,

$$Q(F, F) = \mathcal{L} F + R(F). \tag{2.8}$$

Proof. Put $B_\infty = \int_0^\infty \sqrt{s}\, G(s)\, ds = m(F)/(2\pi)$. For $r \geq 8$, subtraction of the uncorrected leading radial part from (2.2)–(2.3), after division by $8\pi^2$, leaves

$$A_e^2 - \frac{2(B_\infty - B_e)A_e}{r} + \frac{C_e}{r} + F(r)\left[r(B_\infty - B_e) - \frac{E_e}{3r} - D_e - \frac{e}{3}A_e\right]. \tag{2.9}$$

For $N = \|F\|_6$ and $e \geq 1$, elementary integration gives

$$|A_e| \leq CNe^{-2}, \quad |B_\infty - B_e| \leq CNe^{-3/2}, \quad |D_e| \leq CNe^{-1}, \quad |E_e| \leq CN.$$

In C_e, if $s \leq e/2$, then $t \geq e/2$, and

$$\int_{e-s}^{e} \sqrt{s+t-e}\, dt = \frac{2}{3}s^{3/2}.$$

The two regions with one small input therefore contribute at most $CN^2e^{-7/2}$ to $|C_e|/r$. With both inputs above $e/2$, their product is $O(N^2e^{-6})$, and the integral divided by r contributes $O(N^2e^{-4})$. Hence (2.9) is $O(N^2r^{-7})$. On bounded r, (2.5) and the tail integrals give a uniform CN^2 bound. These estimates also establish the absolute pointwise collision bounds used above. The mass correction contributes only a Gaussian times a bounded quadratic scalar.

For radial profiles $F_1, F_2 \in X_6$, the symmetric polarization is

$$B_R(F_1, F_2) = \frac{R(F_1 + F_2) - R(F_1 - F_2)}{4}.$$

Apply the diagonal bound to $\lambda F_1 \pm \lambda^{-1}F_2$, with $\lambda^2 = \|F_2\|_6/\|F_1\|_6$ when both norms are nonzero. It gives $\|B_R(F_1, F_2)\|_7 \leq 2C\|F_1\|_6\|F_2\|_6$; zero arguments are immediate. Now $R(F) - R(\widetilde{F}) = B_R(F - \widetilde{F}, F + \widetilde{F})$ proves (2.7) and the smooth polynomial mapping properties.

For mass cancellation only the unweighted collision integral is needed. Write $V = (v+v_*)/2$, $v - v_* = dk$, $k \in \mathbb{S}^2$. The joint measure including the collision factor is $dV\, d^3\, dd\, dk\, d\sigma$. Swapping k, σ interchanges the pre- and post-collision velocities and preserves this measure. The absolute loss integral is bounded by

$$4\pi \iint (|v| + |v_*|)|F(v)F(v_*)|\, dv\, dv_* < \infty;$$

the gain has the same bound by this angular swap. Fubini now proves $m(Q(F,F)) = 0$. Together with $m(CF) = 0$, this proves $m(R(F)) = 0$. No energy-weighted collision interchange has been used. □

3 The linear evolution and a flat tail

The evolution on the faster-decaying space

We construct the correction evolution in X_p^0 by limits of compactly supported profiles. The critical r^{-6} tail is absent from this space and will be prescribed separately.

Lemma 3.1. *There is a strongly continuous semigroup $S(t)$ on X_p^0 preserving mass, with $\|S(t)\|_{X_p^0 \to X_p^0} \leq e^{bt}$ for a fixed b. If L is its generator, then $Lu(r) = \mathcal{L}u(r)$ for $u \in D(L)$, and $M \in D(L)$.*

Proof. Start with $4\pi C_{\mathrm{raw}}$. On profiles supported in $[0,R]$, its gain and loss are bounded operators, and the positive variation-of-constants series defines the evolution. The gain uses only larger radii, so zero extension is consistent as R increases. A profile vanishing at R continues to vanish there. Recall that $p = 13/2$. Where $\chi > 0$, $r > 4$, and

$$\frac{(4/r)\int_r^\infty s g_p(s)\,ds}{r g_p(r)} = \frac{4}{p-2}(1+r^{-2}) \le \frac{17}{18} < 1. \tag{3.1}$$

At $\chi = 0$ both gain and loss terms vanish. Thus Ag_p, for $A = \|F\|_p$, is a stationary supersolution bounding both signs of the supported solution. Positivity of the bounded integral series proves contractivity uniformly in R. Density gives a positive contraction semigroup $U(t)$ on X_p^0. Strong continuity and the semigroup identity follow first on supported profiles and then by density.

The perturbation $B = -4\pi\ell(\cdot)M$ is bounded on X_p^0. Set

$$S_0(t) = U(t), \qquad S_{j+1}(t)F = \int_0^t U(t-s)BS_j(s)F\,ds.$$

These are vector-valued integrals for each fixed F, with continuous integrands. Their operator norms are bounded by $\|B\|^j t^j/j!$. The sum is strongly continuous and satisfies $\|S(t)\| \le e^{\|B\|t}$. Uniqueness for the integral equation follows by iterating its norm inequality; it also gives the semigroup identity. This construction does not require operator-norm measurability of $U(t)$.

For supported data,

$$m(U(t)F) = m(F) + 4\pi\int_0^t \ell(U(s)F)\,ds.$$

Boundedness of m, ℓ extends this identity by density. In the perturbed integral equation, the scalar mass derivative is $4\pi\ell(S(t)F) + m(BS(t)F) = 0$. Thus $S(t)$ preserves mass; positivity of $S(t)$ is not asserted.

At each fixed r, evaluation of $C_{\mathrm{raw}}F(r)$ is bounded in the X_p norm. Passing from supported profiles to the integrated equation gives the pointwise integrated identity for S. Taking the generator quotient gives $Lu(r) = \mathcal{L}u(r)$ on $D(L)$. Finally, smooth compact cutoffs M_j of the Gaussian satisfy $M_j \to M$ and $C_{\mathrm{raw}}M_j \to C_{\mathrm{raw}}M$ in X_p^0. Pass to the limit in the integrated raw equation and divide by time; the bounded perturbation then yields $M \in D(L)$. □

The prescribed tail

Let $c = 4\pi$, choose $0 < \varepsilon \le 1$, and define the flat function

$$a(t) = \begin{cases} \varepsilon e^{-1/t^2}, & t > 0, \\ 0, & t \le 0. \end{cases}$$

For $r > 0$ and $t \ge 0$, set

$$h(t,r) = \frac{c^5}{6r}\int_0^t a(s)(t-s)^4 e^{-cr(t-s)}\,ds, \quad \widetilde{H} = \chi h, \quad \mu = m(\widetilde{H}), \quad H = \widetilde{H} - \mu M. \tag{3.2}$$

Thus $m(H) = 0$ and $\widetilde{H} \ge 0$.

Lemma 3.2. *The map H is C^∞ in time into X_6, with all time jets zero at $t = 0$. For sufficiently small bounded times,*

$$\|H(t)\|_6 \le Ca(t), \qquad 0 \le \mu(t) \le Ct^3 a(t), \qquad \lim_{r\to\infty} r^6 H(t,r) = 4a(t). \tag{3.3}$$

The defect

$$J = H_t - \mathcal{L}H \tag{3.4}$$

is C^∞ in time into X_p^0, has zero mass and flat initial jets, and satisfies $\|J(t)\|_p \le Ca(t)$. Here $\mathcal{L}H$ is the pointwise expression; no assertion that $H \in X_p^0$ or $H \in D(L)$ is made.

Proof. For $k(u,r) = c^5u^4e^{-cru}/(6r)$,

$$\int_r^\infty sk(u,s)\,ds = \frac{c^4}{6}u^3e^{-cru}, \qquad \partial_u k = cC_0k.$$

Integration by parts in $h = \int_0^\infty a(t-u)k(u,r)\,du$ therefore gives $h_t = cC_0h$; the boundary at $u = 0$ vanishes. After $z = cru$,

$$r^6h(t,r) = \frac{1}{6}\int_0^\infty z^4e^{-z}a(t - z/(cr))\,dz \longrightarrow 4a(t), \tag{3.5}$$

by dominated convergence and $\int_0^\infty z^4e^{-z}\,dz = 24$.

For $r \ge 4$, the norm of this positive kernel as a map from bounded scalar time functions to X_6 is at most $4(17/16)^3$. Apply that bound to the uniform scalar difference-quotient remainders of a. It proves every X_6-valued time derivative and its flat initial trace. This uses the supremum of the integrated kernel, not the generally nonintegrable integral of its supremum. Since a is increasing, it also gives the first bound in (3.3). Integrating the positive kernel for mass and using $\int_0^\infty re^{-cru}\,dr = (cu)^{-2}$ gives the bound on μ. The Gaussian correction has no effect on the limit.

The expression

$$\widetilde{H}_t - cC_{\mathrm{raw}}\widetilde{H} = c\chi[C_0h - C_0(\chi h)]$$

is supported in $4 \le r \le 8$. All other terms in J are scalar multiples of M or $C_{\mathrm{raw}}M$, both rapidly decreasing. Their coefficients are smooth by the kernel bound and bounded moment functionals. For the amplitude bound, use $h_t = cC_0h$ to obtain

$$|\mu'(t)| \le C\int_4^\infty r^3h(t,r)\,dr \le Ca(t).$$

The same bound controls the compactly supported commutator term, μ, and $\ell(H)$. In particular no bound on a'/a is needed. All these expressions are flat, lie in X_p^0, and have zero mass because $m(H) = m(\mathcal{L}H) = 0$. □

4 A smooth signed solution

Seek a mass-one profile in the form

$$F = M + H + w, \qquad w(0) = 0, \qquad m(w) = 0, \qquad w(t) \in X_p^0.$$

Since $Q(M,M) = 0$, Lemma 2.1 gives $R(M) = -\mathcal{L}M = -LM$. Consequently the equation for the correction is

$$w_t = Lw + \Phi(t,w), \qquad \Phi(t,z) = R(M + H(t) + z) - R(M) - J(t). \tag{4.1}$$

Its mild form is

$$w(t) = \int_0^t S(t-s)\Phi(s,w(s))\,ds. \tag{4.2}$$

All constants below are uniform for $0 < \varepsilon \le 1$ on a fixed sufficiently short interval; they may depend on p, χ, M.

Proposition 4.1. *There is an interval $[0,\tau]$ on which (4.2) has a mass-zero solution, with*

$$\|w(t)\|_p \le C\int_0^t a(s)\,ds \le Cta(t). \tag{4.3}$$

For every $t_* < \tau$, the solution is C^∞ in time into X_p^0 on $[0,t_*]$, all initial time jets vanish, and $w(t) \in D(L)$. The profile $F = M + H + w$ has mass one and satisfies $F_t = Q(F,F)$ pointwise.

Proof. In a fixed closed ball of $C([0,\tau];X_p^0)$, the right side of (4.2) has Lipschitz constant at most $e^{b\tau}C\tau$. For small τ it is a contraction mapping the ball into itself. The mass-zero constraint defines a closed subspace and is preserved by both the semigroup and the forcing. The Banach fixed point theorem therefore applies. The same estimates give

$$\|w(t)\|_p \le C\int_0^t a(s)\,ds + C\int_0^t \|w(s)\|_p\,ds.$$

Gronwall, or iteration of this inequality and the factorial remainder, proves (4.3).

We give the time regularity argument because the generator is unbounded. The map Φ is smooth into X_p^0, quadratic in z, and every pure time derivative of $\Phi(t,0)$ is zero at zero. For $h > 0$, the semigroup property gives, for $0 \le t \le \tau - h$,

$$\frac{w(t+h)-w(t)}{h} = S(t)\frac{w(h)}{h} + \int_0^t S(t-s)\frac{\Phi(s+h,w(s+h))-\Phi(s,w(s))}{h}\,ds. \tag{4.4}$$

The first term tends uniformly to zero by (4.3). Taylor's formula for Φ and the Lipschitz integral estimate first bound these quotients and then show that they converge uniformly on each strictly smaller time interval to the unique solution

$$y(t) = \int_0^t S(t-s)[\partial_t\Phi(s,w(s)) + D_z\Phi(s,w(s))y(s)]\,ds.$$

For clarity, the Taylor remainder tends uniformly to zero because the quotients are uniformly bounded and the derivatives of the polynomial map Φ are uniformly continuous on the range under consideration. Subtracting the equation for y from (4.4) and iterating the integral inequality proves convergence without extraction. Left derivatives agree at positive times. Hence w is C^1, $w'(0) = 0$, and w' satisfies the differentiated mild equation.

Repeat the argument for the differentiated system. At each order the highest derivative enters linearly through $D_z\Phi$; the other terms use previously obtained derivatives. Every initial compatibility value is zero because all earlier jets and all pure time jets of $\Phi(t,0)$ vanish. This proves all time derivatives, on the same strictly interior interval, with flat initial traces.

Finally (4.2) implies

$$\frac{S(h)w(t)-w(t)}{h} = \frac{w(t+h)-w(t)}{h} - \frac{1}{h}\int_t^{t+h} S(t+h-s)\Phi(s,w(s))\,ds.$$

The right side tends in X_p^0 to $w'(t) - \Phi(t,w(t))$. This is exactly the generator-domain condition. Every forward quotient is taken below τ; in particular it is valid at the final displayed endpoint $t_* < \tau$. Using $Lw = \mathcal{L}w$, the definition of J, and (2.8) gives $F_t = Q(F,F)$ at each finite r, including zero by the continuous formulas. The mass is one by construction. □

Proposition 4.2. *For every $t_* < \tau$, the profile $F(t,v)$ is jointly C^∞ on $[0,t_*]\times\mathbb{R}^3$, including the velocity origin.*

Proof. We already have $F \in C_t^\infty X_6$. Recall the energy coordinate $e = |v|^2$ and its profile $G(t,e) = F(t,\sqrt{e})$. Therefore

$$A(t) = \int_0^\infty G(t,e)\,de, \qquad D(t) = \int_0^\infty eG(t,e)\,de$$

are smooth scalar functions, since the two functionals are bounded on X_6. This argument uses no higher energy moments. Fix $E < \infty$. In (2.2)–(2.3), replace

$$A_e = A(t) - \int_0^e G(t,s)\,ds, \qquad D_e = D(t) - \int_0^e sG(t,s)\,ds$$

and use (2.5). The equation on $[0,E]$ is now an ODE on $C([0,E])$ with the known smooth coefficients A, D. Its vector field $\mathcal{V}_E(t,G)$ is locally Lipschitz on every Banach space $C^k([0,E])$ with its usual norm. The kernels in (2.5) have fixed compact integration domains and are integrable; differentiating $G(eb)$ only introduces factors $b \in [0,1]$. The same holds for the other local integrals.

For $k \geq 1$, differentiation also gives the tame bound

$$\|\mathcal{V}_E(t,G)\|_{C^k} \leq C_{k,E}\big(|A|,|D|,\|G\|_{C^{k-1}}\big)\big(1 + \|G\|_{C^k}\big) \tag{4.5}$$

on bounded lower-norm sets. A derivative of order k in any product is multiplied by an undifferentiated bounded factor; all other terms involve orders at most $k-1$. Polynomial factors in e are bounded on the fixed interval.

Starting from $G(0,e) = \pi^{-3/2}e^{-e}$, the local contraction in C^k constructs a C^k-valued solution. Uniqueness in C^0 identifies it with the profile already obtained. Inductively, (4.5) and Gronwall bound its C^k norm throughout the existing time interval. This prevents earlier termination: the vector field is bounded in C^k, so a solution approaching a putative termination time has a C^k limit and the local contraction restarts it. Thus every k holds on the same interval. The ODE, with its smooth coefficients, then gives all mixed time and energy derivatives.

Finally $F(t,v) = G(t,|v|^2)$ is smooth at $v = 0$ as well. For a one-sided C^1 energy profile, $\partial_{v_i} G(|v|^2) = 2v_i G'(|v|^2)$, including at zero since $[G(h^2) - G(0)]/h = O(h)$. Apply this identity successively to all energy derivatives and use the polynomial product rule. No extension to negative energies is required. □

5 Positivity and the rapid-decay consequence

Positivity of the semigroup $S(t)$ was not established in Lemma 3.1. We prove nonnegativity for the nonlinear solution itself. Shorten $t_* < \tau$ whenever required below. With $\mu \leq 1/2$, the baseline

$$P = M + H = (1-\mu)M + \widetilde{H}$$

is nonnegative. For $n = F_- = \max(-F, 0)$ and $N(t) = \|n(t)\|_p$, (4.3) gives

$$0 \leq n \leq |w|, \qquad N(t) \leq Cta(t). \tag{5.1}$$

Finite support of the negative part

There are fixed $K_0, k_0 > 0$ such that

$$\widetilde{H}(t,r) \geq k_0 a(t) r^{-6} \quad \text{when } r \geq K_0 t^{-3}. \tag{5.2}$$

To see this, restrict $u = t - s$ in (3.2) to $[1/(cr), 2/(cr)]$. For $u \le t/2$,

$$\log \frac{a(t-u)}{a(t)} = -\frac{2tu - u^2}{t^2(t-u)^2} \ge -\frac{8u}{t^3}.$$

For $r \ge K_0 t^{-3}$ the restricted interval lies below $t/2$, and the cutoff equals one after a fixed shortening of time. The change $z = cru$ bounds $r^6 h/a(t)$ below by

$$\frac{e^{-16/(cK_0)}}{6} \int_1^2 z^4 e^{-z}\, dz > 0.$$

On the same region the ratio of $|w|$ to the lower bound (5.2) is at most

$$Ctr^{6-p} \le CK_0^{6-p} t^{1+3(p-6)} \longrightarrow 0.$$

Hence $F > 0$ for $r \ge R(t) := K_0 t^{-3}$. The negative part has compact support at every positive time, with one common compact support on any compact positive-time interval. Its weighted norm is continuous there, and tends to zero at zero by (5.1). The full profile F is not claimed to have finite X_p norm.

A strict mixed gain estimate

Recall the Gaussian loss frequency

$$\nu_M(v) = 4\pi \int_{\mathbb{R}^3} |v - w| M(w)\, dw.$$

It is radial, so we also write $\nu_M(r)$ for $|v| = r$. For every fixed α with $4/(p-2) < \alpha < 1$, there is a constant C_M such that every nonnegative radial $n \le N g_p$ satisfies

$$\langle r \rangle^p\, 2Q^+(M, n)(r) \le \alpha \nu_M(r) N + C_M N. \tag{5.3}$$

To verify it, polarize (2.2). The local mass integral of the Maxwellian and the tail integral of n are

$$B_M(r) = \int_0^{r^2} \sqrt{s}\, G_M(s)\, ds, \qquad A_n(r) = \int_{r^2}^{\infty} G_n(s)\, ds.$$

The principal term is $8\pi^2 (2/r) B_M(r) A_n(r)$, and

$$A_n(r) \le \frac{2N}{p-2} (1 + r^2)^{1-p/2}.$$

After weighting and division by $\nu_M(r)N$, its upper limit at infinity is $4/(p-2)$, since $\nu_M(r)/(8\pi^2 r) \to B_M(\infty)$. Every term containing the Maxwellian tail A_M decays exponentially after polynomial weighting. In the mixed C term, split the Maxwellian energy at $e/2$. Below $e/2$, the other input is at least $e/2$, giving, before division by r, a bound

$$CN(1+e)^{-p/2} \int_0^{\infty} s^{3/2} G_M(s)\, ds.$$

The weighted term is therefore $O(N/r)$. Above $e/2$, the Maxwellian is exponentially small and all remaining integrals are polynomially bounded. On bounded radii the formulas at zero give a uniform $C_M N$ bound. This proves (5.3).

We also need the elementary envelope bound

$$\|Q^+(B, n)\|_\infty \le C \|B\|_6 N \quad (B \ge 0,\ 0 \le n \le N g_p). \tag{5.4}$$

Indeed $g_p \le g_6$, so positivity of the gain bounds its left side pointwise by $\|B\|_6 N Q^+(g_6, g_6)$. The three terms of (2.2) for this envelope are respectively $O(e^{-4}), O(e^{-5/2}), O(e^{-7/2})$ for $e \ge 1$; at zero use (2.5). Thus the envelope gain is bounded.

The compact maximum argument

At a point where $F < 0$, expand $F = F_+ - n$ in the symmetric gain. The nonnegative diagonal terms can be discarded to give

$$n_t \le 2Q^+(F_+, n) - \nu_F n. \tag{5.5}$$

Moreover $F_+ \le M + \widetilde{H} + |w|$, and

$$\nu_F \ge (1-\mu)\nu_M - \nu_{|w|} \ge (1-\delta)\nu_M$$

for any fixed small $\delta > 0$, after shortening time. Here $\nu_M \ge c_0\langle v\rangle$, whereas $\nu_{|w|} \le C\|w\|_p\langle v\rangle$. Fix $\alpha = 17/18$, $\delta = 1/36$. Since $p = 13/2$, $4/(p-2) = 8/9 < \alpha$ and $1 - \delta - \alpha = 1/36 > 0$.

When $N(t) > 0$, its maximum is attained at a negative point with $r \le R(t)$. Apply (5.3) to the Maxwellian contribution in (5.5). For the other gain terms, (5.4) and $\|\widetilde{H} + |w|\|_6 \le Ca(t)$ give, after weighting on this compact interval, at most $Ca(t)t^{-3p}N(t)$. Discarding the strictly negative remaining frequency term yields

$$D^+N(t) \le [C_M + Ca(t)t^{-3p}]N(t), \qquad D^+N(t) = \limsup_{h\downarrow 0} \frac{N(t+h) - N(t)}{h}. \tag{5.6}$$

This use of the derivative of a maximum is local: on a compact positive-time interval all maximizing points lie in a common compact radius interval. A maximizing sequence at $t + h$ has a subsequence converging to a maximizer at t, and uniform integration of the continuous time derivative bounds the upper Dini derivative by the largest derivative at such maximizers. If $N(t) = 0$, then $F \ge 0$; at every zero of F, its equation gives $F_t = Q^+(F, F) \ge 0$. The right derivative of the negative part there is zero, as it is at strictly positive points. Thus (5.6) holds also in the zero-norm case.

The coefficient in (5.6) is integrable at zero, since the flat function $a(t)t^{-3p}$ tends to zero faster than any fixed power. For $0 < \eta < t$, the scalar Dini comparison gives

$$N(t) \le N(\eta) \exp\left(\int_\eta^t [C_M + Ca(s)s^{-3p}]\, ds\right).$$

For completeness, this comparison follows by multiplying N by the integrating factor. A continuous function with nonpositive upper right Dini derivative is nonincreasing: otherwise subtract a small positive linear function and take a minimum before an interval's right endpoint to contradict the strictly negative Dini derivative. Now (5.1) implies $N(\eta) \to 0$, while the exponential remains bounded. Thus $N = 0$ and $F \ge 0$.

The macroscopic bounds and the endpoint

All the preceding reductions select a single positive $t_* < \tau$. Set $f(t, x, v) = F(t, |v|)$. Propositions 4.1 and 4.2, and positivity, give the claimed smooth nonnegative solution on this interval. Its datum is M. Every velocity derivative of M is a polynomial times that Gaussian, and every spatial derivative is zero. The time jets of $H + w$ vanish at zero as well.

There is a fixed $K \ge 1$ with

$$0 \le F(t, r) \le K(1 + r^2)^{-3} \qquad (0 \le t \le t_*).$$

The mass is one, and the energy is at most

$$E_0 = 4\pi K \int_0^\infty r^4(1 + r^2)^{-3}\, dr < \infty.$$

The positive entropy is bounded by $\log K \int F = \log K$. To check that the entropy integral is well defined, for $b > 0, z \geq 0$ use

$$-z \log z \leq b(1 + |v|^2)z + e^{-1-b(1+|v|^2)}.$$

The scalar inequality follows by maximizing $-z \log z - cz$ with $c = b(1 + |v|^2)$. Its Gaussian remainder is integrable, so the negative entropy part is uniformly finite as well. No entropy dissipation law is required.

Finally the Gaussian and $w \in X_p^0$, $p > 6$, have zero r^6 tail. Equations (3.3) and (3.5) give (1.5). This proves Theorem 1.1.

Corollary 5.1. *For the full-sphere hard-sphere equation (1.1) on $\mathbb{T}^3 \times \mathbb{R}^3$, joint pointwise smoothness, initially Schwartz data with Schwartz derivatives, and the uniform local bounds (1.2) do not imply an extension with rapid velocity decay at the endpoint. More precisely, for any $0 < T < t_*$, the restriction of the solution in Theorem 1.1 to $[0, T)$ satisfies these hypotheses with $m_0 = M_0 = 1$ and finite E_0, H_0, but every continuous extension to T has*

$$\lim_{r \to \infty} r^6 f(T, x, r\omega) = 4\varepsilon e^{-1/T^2} > 0.$$

Proof. At each finite velocity an extension must have the existing smooth endpoint value, so its value at T is precisely the constructed profile. That profile has the displayed tail and cannot be rapidly decreasing. □

Here t_*, ε are those of Theorem 1.1, T is any strictly interior positive time, r is velocity radius, $\omega \in \mathbb{S}^2$ is a unit direction, and $x \in \mathbb{T}^3$. The solution already extends smoothly in X_6 across T. Thus the failed conclusion is rapid velocity decay. The example does not resolve the continuation question in a conservative, energy-admissible, or positive-time Schwartz solution class.

AI disclosure. OpenAI Codex was used for proof development, calculation checks, and drafting. The mathematical claims and the limitations of their scope are stated in the text. One fresh-context AI reader and one fresh-context AI referee provided feedback on a preceding draft; the Codex writing assistant then revised it once.

References

[1] L. Silvestre, *Regularity estimates and open problems in kinetic equations*, arXiv:2204.06401v1 (2022). arXiv:2204.06401v1.

[2] X. Lu and B. Wennberg, *Solutions with increasing energy for the spatially homogeneous Boltzmann equation*, Nonlinear Analysis: Real World Applications **3** (2002), no. 2, 243–258. doi:10.1016/S1468-1218(01)00026-8.

[3] G. Basile, D. Benedetto, L. Bertini, and E. Caglioti, *Asymptotic probability of energy increasing solutions to the homogeneous Boltzmann equation*, arXiv:2202.07311v2 (2022). arXiv:2202.07311v2.